ARJUNA JEE 2027

Lecture -01 | Mathematics

TRIGONOMETRIC FUNCTIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Log & Exponential Series**

B **Introduction to Trigonometry (Class-X)**

C **Question Practice**

D



Question Practice

Acidic Question

If a, b, c are +ve numbers & $a + 2b + 3c = 7$ then find maximum value of :

(i) (ab^2c^3) (ii) $(a^2b^3c^4)$ (iii) (ab^2c)

$ab^2c \rightarrow abbc$

$a+2b+3c = 7.$

$a, b, b, 3c \rightarrow AM \geq GM$

$$\frac{a+2b+3c}{4} \geq (3ab^2c)^{\frac{1}{4}}$$

$$\frac{7}{4}^4 \geq (3ab^2c)$$

$$\frac{7^4}{3\times 4^4} \geq ab^2c$$

When: $a = b = 3c$

$a + 2a + a = 7$

$a = \frac{7}{4}, b = \frac{7}{4}, c = \frac{7}{12}$

Question

If $4x + 3y = 14$, then maximum value of x^4y^3 is 'a' which occurs at $x = b$ and $y = c$. Value of $(a + b + c)$ is (where $x, y \in R^+$)

A 126

B 128

C 130

D 132

QIBY - A

Question [JEE (Adv.)-2020, Paper-1]

$m = \log_3(3^4) = 4$

80%

Let m be the minimum possible value of $\log_3(3^{y_1} + 3^{y_2} + 3^{y_3})$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^3)$ is _____.

$3^{y_1}, 3^{y_2}, 3^{y_3} \rightarrow$ AM ≥ GM

$$\frac{3^{y_1} + 3^{y_2} + 3^{y_3}}{3} \geq \left(3^{y_1+y_2+y_3}\right)^{\frac{1}{3}}$$

$$3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3^4$$

$$\frac{x_1 + x_2 + x_3}{3} \geq (x_1 x_2 x_3)^{\frac{1}{3}}$$

$$3^3 \geq x_1 x_2 x_3$$

$$\log_3(x_1 x_2 x_3)\big|_{\max} = \log_3(3^3) = 3 = M$$

$\log_2(4^3) + \log_3(3^3)$

$6 + 3 = 9$

Ans.

ACIDIC QUESTION

If $a_1, a_2, a_3 \ldots a_n \in R^+$ and $a_1 \cdot a_2 \cdot a_3 \ldots \ldots a_n = 1$ then show that minimum value of : $(1 + a_1 + a_1^2)(1 + a_2 + a_2^2) \ldots . (1 + a_n + a_n^2)$ is 3^n.



EXPONENTIAL & LOG SERIES



FACTORIAL

1. $n! = 1.2.3 \ldots\ldots n, n \in N$ → (T)
2. $0! = 1$ → (T)
3. $(2n)! = 2(n!)$ → (F)
4. $(2n)! = 2n \cdot (2n-1)!$ → (T)
5. $(n+13)! = (n+13) \cdot (n+12)!$ → (T)
6. **Factorial is only defined for whole numbers.** → (T)

$18! = 18 \times 17!$

Bumper QUESTION

Find the sum of following series :

(i) $S = \sum_{n=1}^{100} n(n!)$ (ii) $S = \sum_{n=1}^{50} \frac{n}{(n+1)!}$ (iii) $S = \sum_{r=1}^{100} (r^2+1) \cdot r!$

1 4 18

$1.1! + 2.2! + 3.3! + \ldots + 100.100!$

$T_n = n(n!) = (n+1-1)n! = (n+1).n! - n! = (n+1)! - n!$

$T_1 = 2! - 1!$

$T_2 = 3! - 2!$

$T_3 = 4! - 3!$

$\vdots$

$T_{100} = 101! - 100!$

$\Big\} \; f(n) = (101! - 1)$

Bumper QUESTION

Find the sum of following series :

(i) $S = \sum_{n=1}^{100} n(n!)$

(ii) $S = \sum_{n=1}^{50} \frac{n}{(n+1)!}$

(iii) $S = \sum_{r=1}^{100} (r^2+1) \cdot r!$

$T_1 = \frac{1}{1!} - \frac{1}{2!}$

$T_2 = \frac{1}{2!} - \frac{1}{3!}$

$T_3 = \frac{1}{3!} - \frac{1}{4!}$

$\vdots$

$T_{50} = \frac{1}{50!} - \frac{1}{51!}$

$T_n = \frac{(n+1)-1}{(n+1)!} = \frac{n+1}{(n+1)!} - \frac{1}{(n+1)!}$

$T_n = \frac{1}{n!} - \frac{1}{(n+1)!}$

$\frac{n+1}{(n+1) \cdot n!} = \frac{1}{n!}$

$f(n) = \left(\frac{1}{1!} - \frac{1}{51!}\right)$

Bumper QUESTION

RAM

Find the sum of following series :

(i) $S = \sum_{n=1}^{100} n(n!)$

(ii) $S = \sum_{n=1}^{50} \frac{n}{(n+1)!}$



(iii) $S = \sum_{r=1}^{100} (r^2 + 1) \cdot r!$

$r^2+1 = r^2 + 1^2 + 2r(1) - 2r(1)$

$= (r+1)^2 - 2r$

$\left((r+1)^2 - 2r\right) \cdot r!$

$(r+1).(r+1).r! - 2.r.r!$

$(r+1).(r+1)! - r.r! - r.r!$

$$T_r = (r+1).(r+1)! - r.r! - r.r!$$

$$T_1 = 2.2! - 1.1! - 1.1!$$

$$T_2 = 3.3! - 2.2! - 2.2!$$

$$T_3 = 4.4! - 3.3! - 3.3!$$

$$\vdots$$

$$T_{100} = 101.101! - 100.100! - 100.100!$$

$$101.101! - 1.1! - (1.1! + 2.2! + 3.3! + \ldots + 100.100!)$$

$$101.101! - 1 - (101! - 1) = 101.101! - 1 - 101! + 1 = (101-1)101!$$

fnf

$$100.(101!)$$

M-2

RAM

$$\# \sum_{r=1}^{100} (r^2+1)\cdot r!$$

$+r-r$

$$(r^2+r+1-r)\, r!$$

$$(r^2+r)\cdot r! + (1-r)\, r!$$

$$r(r+1)\cdot r! - (r-1)\cdot r!$$

$$T_r = r\cdot(r+1)! - (r-1)\cdot r!$$

$T_1 = 1\cdot 2! - 0\cdot 0!$

$T_2 = 2\cdot 3! - 1\cdot 2!$

$T_3 = 3\cdot 4! - 2\cdot 3!$

$T_4 = 4\cdot 5! - 3\cdot 4!$

$\vdots$

$T_{100} = 100\cdot 101! - 99\cdot 100!$

$\text{fof} = 100\cdot 101!$

Question

$\sum_{r=1}^{99} r!\left(r^2+r+1\right)$ is equal to



A $102! - 100\ !$

B $100(100!) - 1$

C $99(100!) - 1$

D $100(99!) - 1$



EXPONENTIAL SERIES

$$e^x = \frac{1}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\dots \infty \quad , \; x \in Q.$$

RAR

$$\# \; e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

RAR

$$e^x = \frac{1}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty \quad , \; x \in Q.$$

$x = 1$

$$e^1 = \left(1 + \frac{1}{1!}\right) + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \infty$$

$2.73\dots$ $\quad$ $2 + 0.738\dots$

$x = 2$

$$e^2 = 1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \dots \infty$$

$x = -1$

$$\frac{1}{e} = e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots \infty$$

RAR

$$e^x = \frac{1}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty \quad , \; x \in Q.$$

$x \to -x$

$$e^{-x} = \frac{1}{0!} - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots \infty$$

add

$$\frac{e^x + e^{-x}}{2} = \left(\frac{1}{0!} + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots \infty\right)$$

$x = 1$

$$\frac{e^1 + e^{-1}}{2} = \frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \infty$$

RAR

$$e^x = \frac{1}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty \quad , \; x \in Q.$$

$x \to -x$

$$e^{-x} = \frac{1}{0!} - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots \infty$$

$- \quad + \quad - \quad + \quad -$

Sub.

$$\frac{e^x - e^{-x}}{2} = \left(\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty\right) \xrightarrow{x=1}$$

$$\frac{e^1 - e^{-1}}{2} = \frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \infty$$

RAPID FIRE ROUND

A. $\frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots\ldots\infty = e^1$

B. $1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\ldots\ldots\infty = e^{-1} = \frac{1}{e}$

C. $\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}\ldots\ldots\infty = e-1$.

D. $\frac{2}{1!}+\frac{4}{2!}+\frac{8}{3!}+\frac{16}{4!}\ldots\ldots\infty = e^2-1$.

RAPID FIRE ROUND

E. $\frac{1}{0!}+\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+\ldots\ldots\infty = \frac{e^1+e^{-1}}{2}$

F. $\frac{1}{1!}+\frac{1}{3!}+\frac{1}{5!}+\frac{1}{7!}+\ldots\ldots\infty = \frac{e^1-e^{-1}}{2}$

G. $\frac{1}{1!}+\frac{2}{2!}+\frac{3}{3!}+\frac{4}{4!}\ldots\ldots\infty = \sum_{n=1}^{\infty}\frac{n}{n!} = \sum_{n=1}^{\infty}\frac{1}{(n-1)!} = \frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\ldots\infty = e$

$T_n = \frac{n}{n!} = \frac{n}{n\cdot(n-1)!} = \frac{1}{(n-1)!}$

EXPONENTIAL SERIES

RAR

$$e^x = \frac{1}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty \quad , \; x \in Q.$$

$x \to x \ln 2$

$$2^x = e^{x \ln 2} = \frac{1}{0!} + \frac{\ln 2 \cdot x}{1!} + \frac{(\ln 2)^2 x^2}{2!} + \frac{(\ln 2)^3 \cdot x^3}{3!} + \dots \infty$$

$$(v_1)^{v_2} = e^{v_2 \ln v_1}$$

$$2^3 = e^{3\ln 2} = (e^{\ln 2})^3 = (e^{\log_e 2})^3 = (2^{\underbrace{\log_e e}_{1}})^3$$

$$= 2^3$$

$$5^4 = e^{4\ln 5}$$

IMPORTANT NOTE

$$a^x = e^{x\ln a} = 1 + \frac{x}{1!}\ln a + \frac{x^2}{2!}(\ln a)^2 + \frac{x^3}{3!}(\ln a)^3 + \ldots\ldots\infty$$

no need to RAR

Acidic Question

Find the sum of following series :

$$\sum_{n=1}^{\infty} \frac{n^2}{n!} = \frac{1^2}{1!} + \frac{2^2}{2!} + \frac{3^2}{3!} + \frac{4^2}{4!} + \ldots\infty = 2e$$



$$T_n = \frac{n^2}{n!} = \frac{n \times n}{n \cdot (n-1)!} = \frac{(n-1)+1}{(n-1)\cdot(n-2)!} = \frac{(n-1)}{(n-1)(n-2)!} + \frac{1}{(n-1)(n-2)!}$$

$n \neq 1$

$$T_n = \frac{1}{(n-2)!} + \frac{1}{(n-1)!}$$

$$T_n = \frac{1}{(n-2)!_0} + \frac{1}{(n-1)!_0}$$

$T_1 = \frac{1}{(-1)!_0} + \frac{1}{0!_0} \to 0$

$T_2 = \frac{1}{0!_0} + \frac{1}{1!_0}$

$T_3 = \frac{1}{1!_0} + \frac{1}{2!_0}$

$T_4 = \frac{1}{2!_0} + \frac{1}{3!_0}$

$\vdots$

$\infty \to e \qquad \infty \to e$

$\#\ \frac{1}{(-n)!_0} = 0\ ,\ n \in N.$

$\frac{1}{(-1)!_0} = \frac{1}{(-2)!_0} = \frac{1}{(-3)!_0} = \frac{1}{(-4)!_0} = \ldots = 0$

Acidic Question

Find the sum of following series :

$$\sum_{n=1}^{\infty} \frac{n^3}{n!} = \frac{1^3}{1!} + \frac{2^3}{2!} + \frac{3^3}{3!} + \frac{4^3}{4!} + \ldots \infty$$



Acidic Question

Find the sum of following series :

$$\frac{1}{0!}+\left(\frac{1+2}{1!}\right)+\left(\frac{1+2+3}{2!}\right)+\left(\frac{1+2+3+4}{3!}\right)+\ldots\ldots\infty$$

$$T_n = \frac{1+2+3+\ldots+n}{(n-1)!} = \frac{n(n+1)}{2\times(n-1)!}$$

$$(n-1)\cdot(n-2)!$$



LOGARITHMIC SERIES

$$\ln(1-x) = \log_e(1-x) = -\frac{x^1}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \cdots\cdots \infty \quad , \quad -1 \le x < 1$$

$$x \to (-x)$$

$$\ln(1+x) = \log_e(1+x) = \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots\cdots \infty \quad , \quad -1 < x \le 1$$

$$\ln(1+x) = \log_e(1+x) = \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots\cdots \infty$$

$x = 1$

$$\ln 2 = \boxed{1 - \frac{1}{2}} + \boxed{\frac{1}{3} - \frac{1}{4}} + \boxed{\frac{1}{5} - \frac{1}{6}} + \boxed{\cdots} \infty$$

$$\ln 2 = \frac{1}{1\times 2} + \frac{1}{3\times 4} + \frac{1}{5\times 6} + \frac{1}{7\times 8} + \cdots \infty$$



LOGARITHMIC SERIES

$$\ln(1-x) = -\frac{x^1}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \ldots\ldots\infty, \qquad -1 \le x < 1$$

$$\ln(1+x) = \frac{x^1}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} \ldots\ldots\infty, \qquad -1 < x \le 1$$

Question

$$1+\frac{2}{3}-\frac{2}{4}+\frac{2}{5}-\ldots\ldots\infty=$$

A $\log_e 3$

B $\log_e 4$

C $\log_e\left(\frac{e}{2}\right)$

D $\log_e\left(\frac{2}{3}\right)$

$$\ln(1+x)=\frac{x^1}{1}-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots\ldots$$

$-1<x\leq 1$

New

Question

Evaluate the sum of the series $\frac{1}{1\cdot 2}-\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}-\frac{1}{4\cdot 5}+\ldots\ldots\infty$



TRIGONOMETRY

1. TRIGONOMETRIC FUNCTIONS (RATIOS & IDENTITIES) (XI)
2. TRIGONOMETRIC EQUATIONS (XI)
3. SOLUTIONS OF TRIANGLE (XI)
4. INVERSE TRIGONOMETRIC FUNCTIONS (XII)

TRIGONOMETRIC FUNCTIONS

(RATIOS & IDENTITIES)

BASIC OF TRIGONOMETRY (CLASS-X)

Point-01: Six Trigonometric Functions

$\sin\theta = \frac{P}{H}$

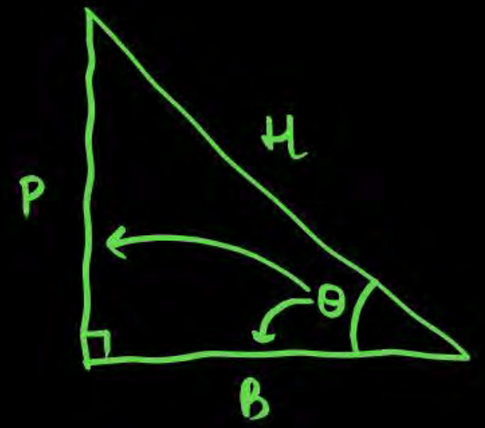
$\cos\theta = \frac{B}{H}$

$\tan\theta = \frac{P}{B} = \frac{\sin\theta}{\cos\theta}$

$\frac{1}{\sin\theta} = \operatorname{cosec}\theta = \frac{H}{P}$

$\frac{1}{\cos\theta} = \sec\theta = \frac{H}{B}$

$\frac{1}{\tan\theta} = \cot\theta = \frac{B}{P} = \frac{\cos\theta}{\sin\theta}$



BASIC OF TRIGONOMETRY (CLASS-X)

Point-02: Trigonometric Identities

(1) $\sin^2\theta + \cos^2\theta = 1$

(2) $\sec^2\theta - \tan^2\theta = 1$

(3) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

(1) $\sin^2\theta + \cos^2\theta = 1$

(2) $\sec^2\theta - \tan^2\theta = 1$

$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

(3) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

OR

$(\operatorname{cosec}\theta - \cot\theta)(\operatorname{cosec}\theta + \cot\theta) = 1$

OR

$(\operatorname{cosec}\theta + \cot\theta) = \dfrac{1}{(\operatorname{cosec}\theta - \cot\theta)}$

$(\sec\theta + \tan\theta) = \dfrac{1}{\sec\theta - \tan\theta}$

OR

$(\sec\theta - \tan\theta) = \dfrac{1}{\sec\theta + \tan\theta}$

Ques: If $\sec\theta - \tan\theta = \alpha$, find $\cos\theta$??

$\sec\theta + \tan\theta = \frac{1}{\alpha}$

add

$2\sec\theta = \alpha + \frac{1}{\alpha} = \frac{\alpha^2+1}{\alpha}$

$2\frac{1}{\cos\theta} = \frac{\alpha^2+1}{\alpha}$

$\frac{2\alpha}{\alpha^2+1} = \cos\theta$ ✓

RAPID FIRE DIBY'S (CLASS-X)

1

Which of the following reduces to unity for $0^\circ < A < 90^\circ$

(1) $(\sec^2 A - 1)\cot^2 A$

(2) $\cos A \operatorname{cosec} A\sqrt{\sec^2 A - 1}$

(3) $(\operatorname{cosec}^2 A - 1)\tan^2 A$

(4) $(1 - \cos^2 A)(1 + \cot^2 A)$

(5) $\sin A \sec A\sqrt{\operatorname{cosec}^2 A - 1}$

(6) $(1 + \tan^2 A)(1 - \sin^2 A)$

(7) $\sec^2 A - \sin^2 A \sec^2 A$

(8) $\frac{1}{1+\sin^2 A} + \frac{1}{1+\operatorname{cosec}^2 A}$

(9) $\frac{\tan^2 A \sin^2 A}{\tan^2 A - \sin^2 A}$

(10) $\frac{\cot^2 A \cos^2 A}{\cot^2 A - \cos^2 A}$

DO IT BY YOURSELF (DIBY)

DIBY-89

If the sum of series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2d)}+\cdots+\frac{1}{(1+9d)(1+10d)}$ is equal to 5, then $50d$ is equal to:

[JEE Mains-2024]

(A) 20 (B) 5 (C) 15 (D) 10

[Ans. : (B)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-94

The sum to 10 terms of the series $\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\cdots$ is

[JEE Mains-2023]

(A) $\frac{59}{111}$ (B) $\frac{55}{111}$ (C) $\frac{56}{111}$ (D) $\frac{58}{111}$

[Ans. : (B)]

DIBY-95

If $a_n=\frac{-2}{4n^2-16n+15}$, then $a_1+a_2+\cdots+a_{25}$ is equal to: [JEE Mains-2023]

(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$

[Ans. : (C)]

DIBY-96

The sum of the series $\frac{1}{1-3\cdot1^2+1^4}+\frac{2}{1-3\cdot2^2+2^4}+\frac{3}{1-3\cdot3^2+3^4}+\ldots$ up to 10 terms is

[JEE Mains-2024]

(A) $\frac{45}{109}$ (B) $-\frac{45}{109}$ (C) $\frac{55}{109}$ (D) $-\frac{55}{109}$

[Ans. : (D)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-97

If the sum of the first 20 terms of the series $\frac{4.1}{4+3.1^2+1^4}+\frac{4.2}{4+3.2^2+2^4}+\frac{4.3}{4+3.3^2+3^4}+\frac{4.4}{4+3.4^2+4^4}+\cdots$ is $\frac{m}{n}$, where m and n are coprime, then $m+n$ is equal to

(A) 423 (B) 420 (C) 421 (D) 422

[Ans. : (C)]

DIBY-98

The sum $\sum_{k=1}^{n}\frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}}$ is equal to

[Ans. : $\frac{\sqrt{n+1}-1}{\sqrt{n+1}}$]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-99

Find the sum $\sum_{n=1}^{\infty} \frac{6^n}{(3^n-2^n)(3^{n+1}-2^{n+1})}$ [Ans. (2)]

DIBY-100

Find the sum of the series $\frac{1}{3^2+1}+\frac{1}{4^2+2}+\frac{1}{5^2+3}+\frac{1}{6^2+4}+\cdots\infty$. [Ans. (13/36)]

DIBY-101

Find the sum $\sum_{n=2}^{\infty} \frac{3n^2+1}{(n^2-1)^3}$ [Ans. (09/16)]

DIBY-102

Let $S=\frac{\sqrt{1}}{1+\sqrt{1}+\sqrt{2}}+\frac{\sqrt{2}}{1+\sqrt{2}+\sqrt{3}}+\frac{\sqrt{3}}{1+\sqrt{3}+\sqrt{4}}+\cdots+\frac{\sqrt{n}}{1+\sqrt{n}+\sqrt{n+1}}=10$.

Then find the value of n. [Ans. (24)]

INEQUALITY

DIBY-103

Minimum value of sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ & a^{10} (where $a > 0$)

[IIT-JEE-2011]

[Ans. 8]

DIBY-104

If $a > 0, b > 0, c > 0$, then show that : $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$ [JEE-1984]

DIBY-105

For positive real number a, b & c show that $\left((1+a)(1+b)(1+c)\right)^7 > 7^7 a^4 b^4 c^4$ [JEE-2004]

DIBY-106

Let $x, y > 0$. If $x^3y^2 = 215$, then the least value of $3x + 2y$ is [JEE Mains-2022]

(A) 30 (B) 32 (C) 36 (D) 40

[Ans. (D)]

INEQUALITY

DIBY-107

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$,, where $a, x \in R$ and $a > 0$, is equal to

[JEE Mains-2021]

(A) $a + \frac{1}{a}$ (B) $2a$ (C) $a + 1$ (D) $2\sqrt{a}$

[Ans. D]

DIBY-108

Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5b^3c^2d$ is 3750β, then the value of β is [JEE Mains-2023]

(A) 110 (B) 108 (C) 90 (D) 55

[Ans. C]

DIBY-109

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}, x > 0$ is 14, then the value of α is equal to

[JEE Mains-2022]

(A) 32 (B) 64 (C) 128 (D) 256

[Ans. (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu)- 4 sittings each of 1 HR

It is never about end result

It's always about Journey

Thank You

#futureIITians